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(54) **SOLAR ROOFING SYSTEM**

(71) Applicant: **GAF Energy LLC**, Parsippany, NJ
(US)

(72) Inventors: **Richard Perkins**, San Jose, CA (US);
Matthew Peterson Grigsby, San
Francisco, CA (US)

(73) Assignee: **GAF Energy LLC**, Parsippany, NJ
(US)

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CPC **H02S 20/25** (2014.12); **E04D 1/20**
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(58) **Field of Classification Search**

CPC Y02B 10/10; Y02E 10/50; H02S 20/25;
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See application file for complete search history.

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Primary Examiner — Brian E Glessner

Assistant Examiner — Daniel J Kenny

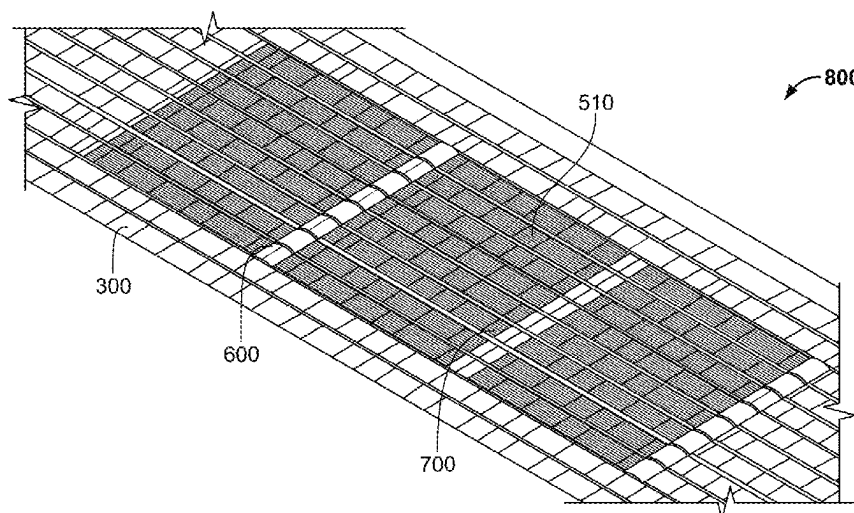
(74) *Attorney, Agent, or Firm* — GREENBERG
TRAURIG, LLP

(57)

ABSTRACT

A system includes a photovoltaic module having photovol-
taic cells, each having a width, and a roofing shingle having
an exposure zone and a headlap zone. A plurality of slots
extends from the exposure zone to the headlap zone and
define tooth portions. A first one of the tooth portions has a
first side defined by a first slot and a second side defined by
a second slot adjacent to the first slot. The first tooth portion
has a first width that is the photovoltaic cell width multiplied
by a first positive integer. A second tooth portion has a first
side defined by a third slot and a second side defined by a
fourth slot adjacent to the third slot. The second tooth
portion has a second width that is the photovoltaic cell width
multiplied by a second positive integer different than the first
positive integer.

20 Claims, 12 Drawing Sheets



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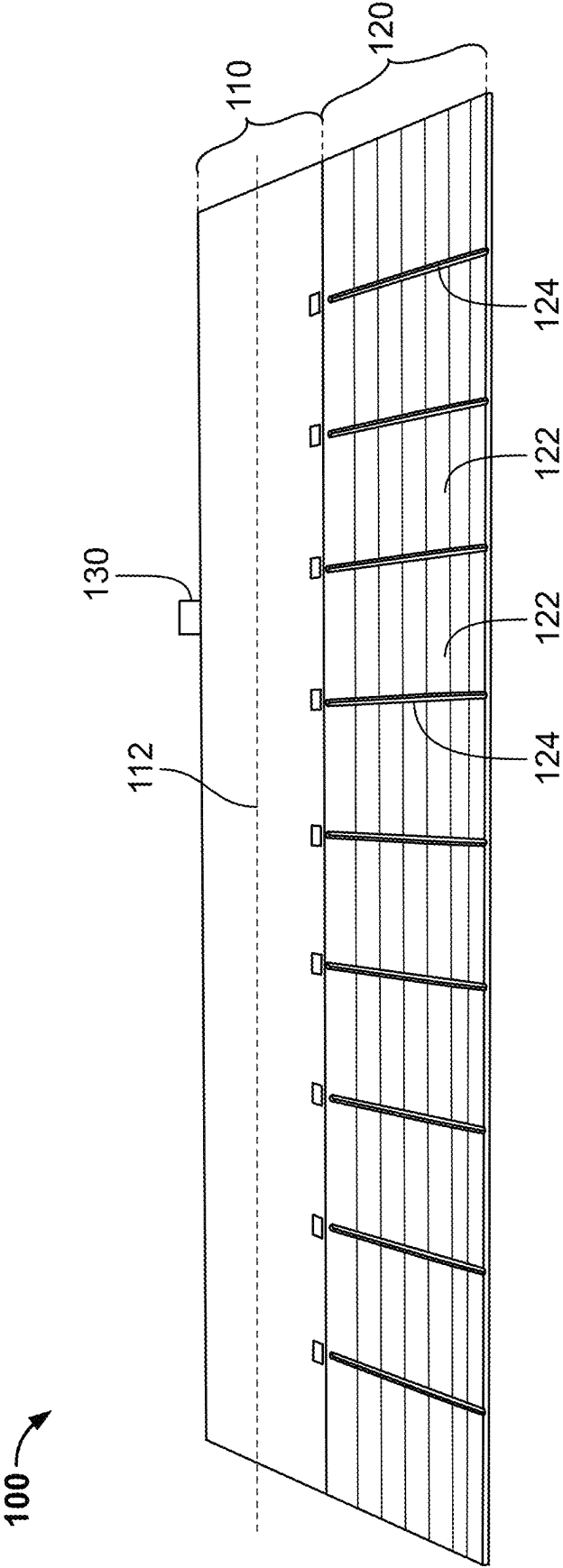


FIG. 1

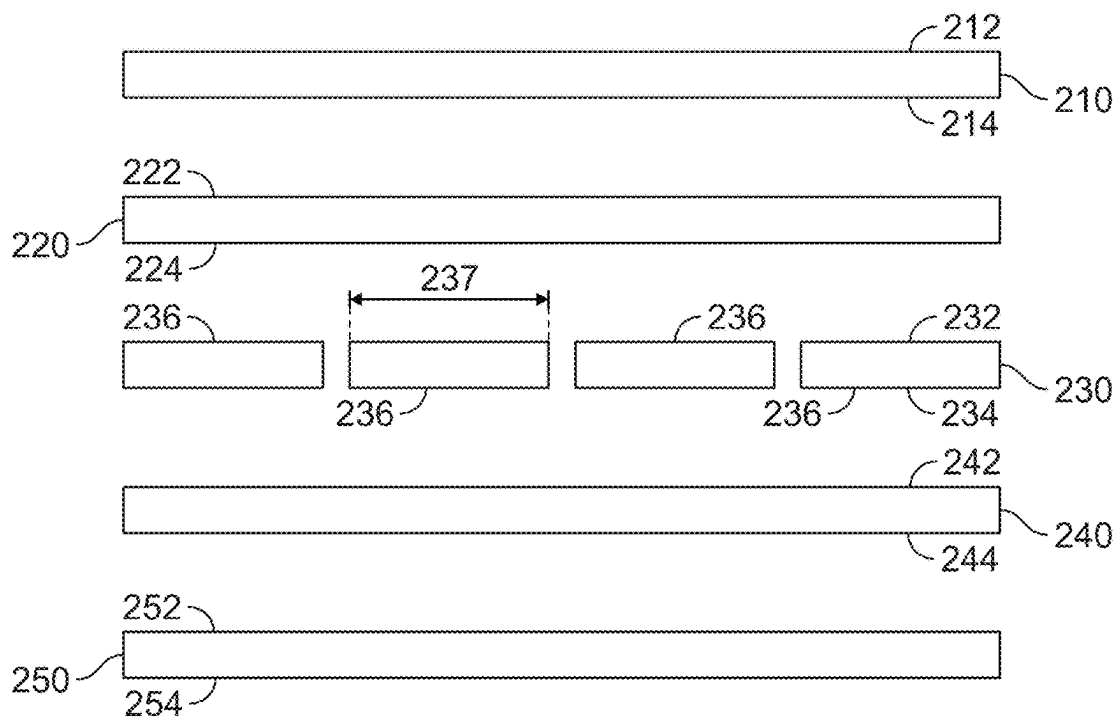


FIG. 2A

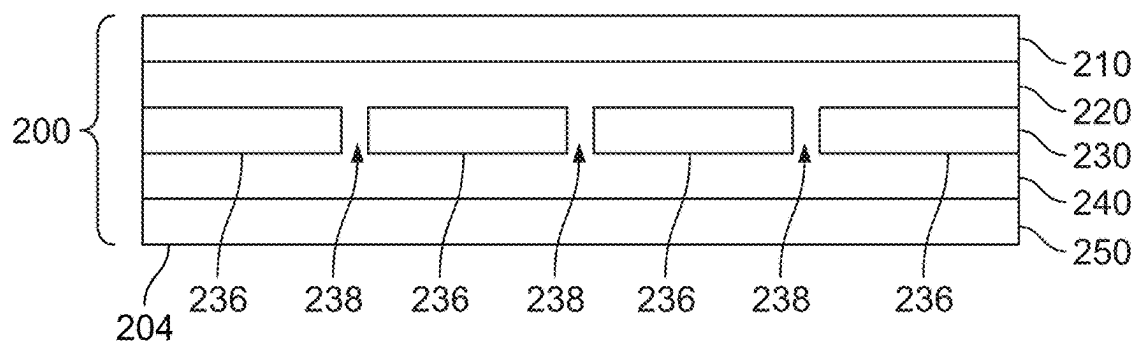


FIG. 2B

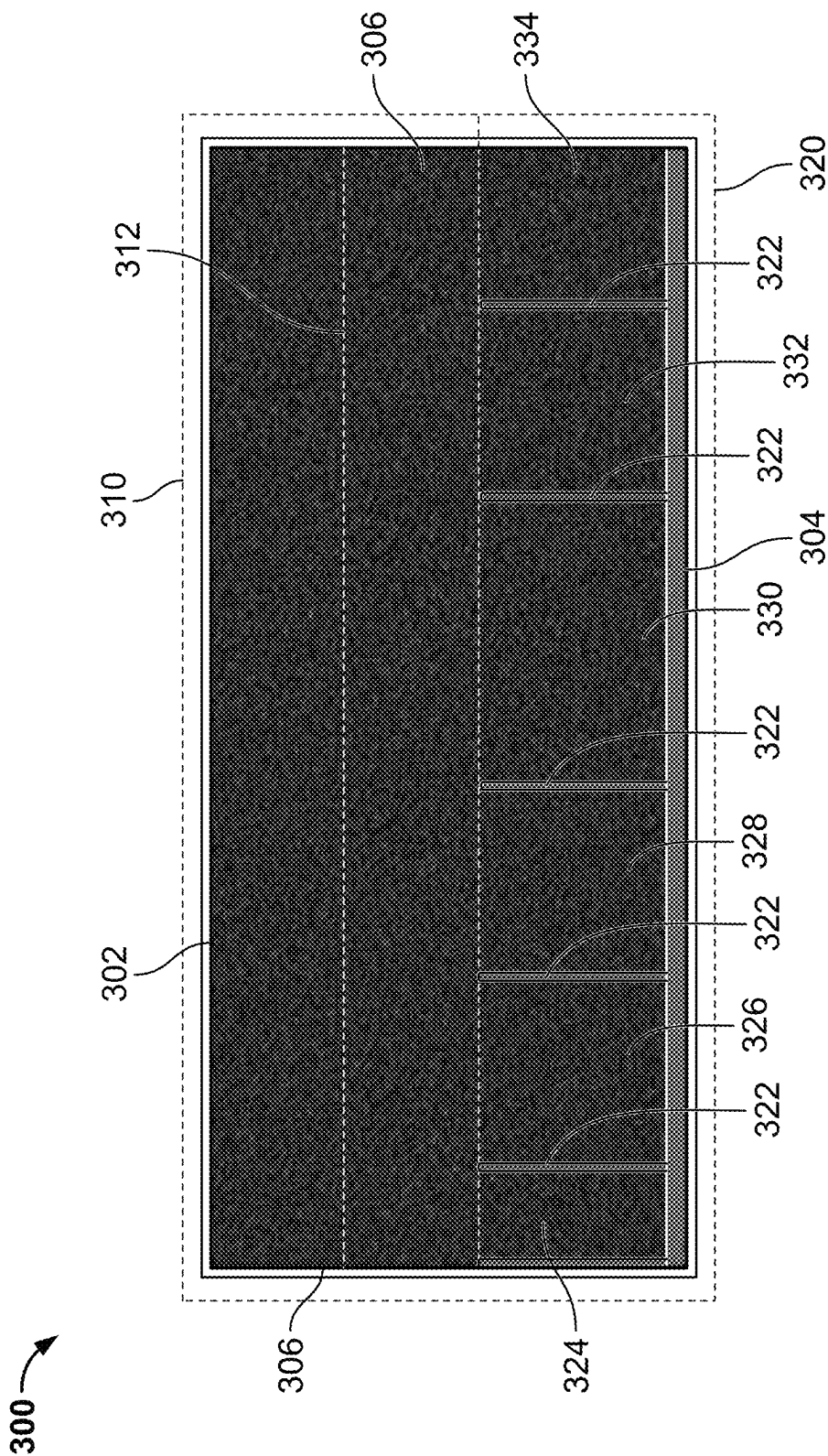


FIG. 3

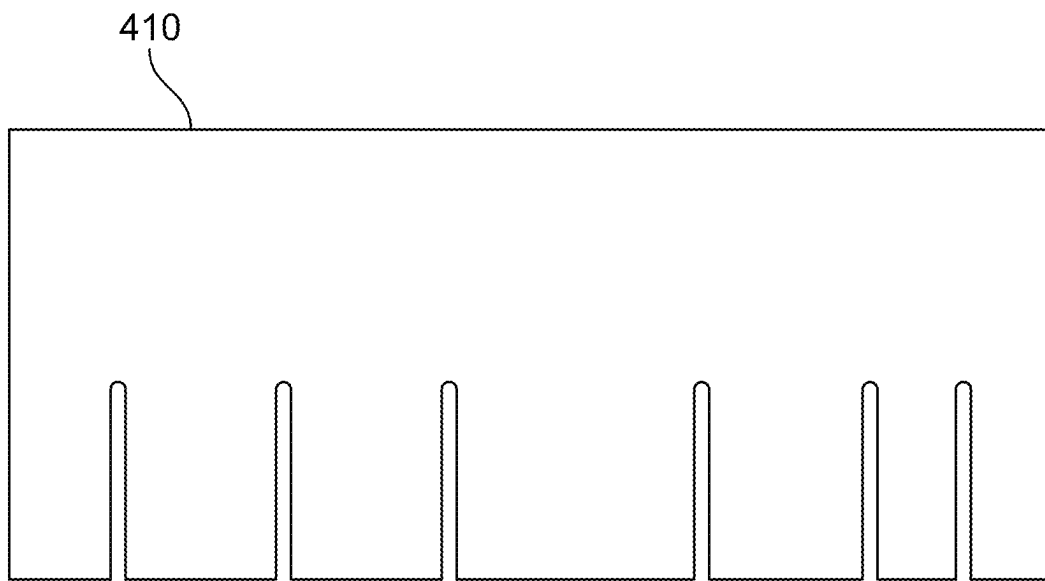


FIG. 4A

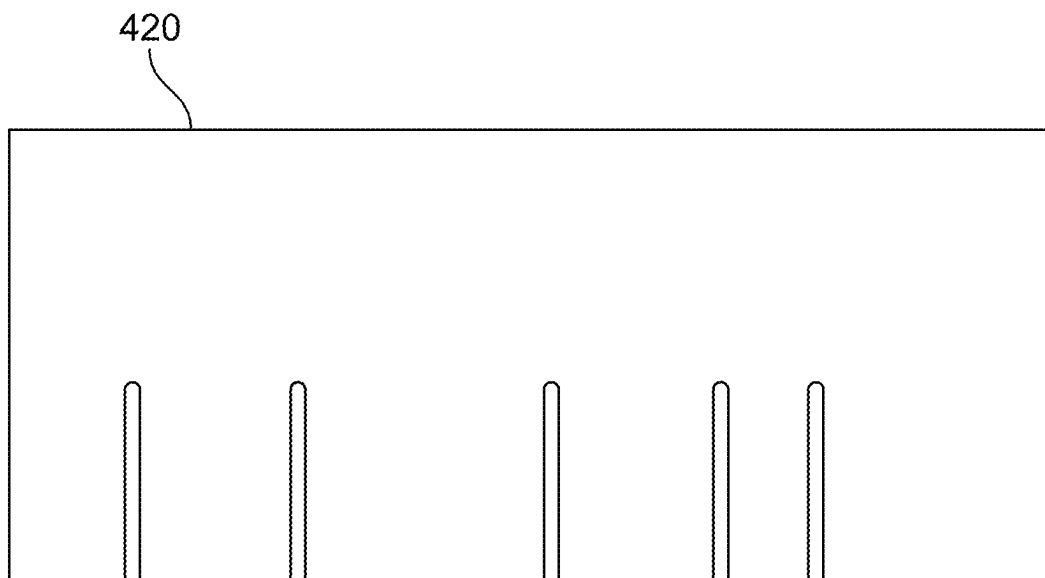


FIG. 4B

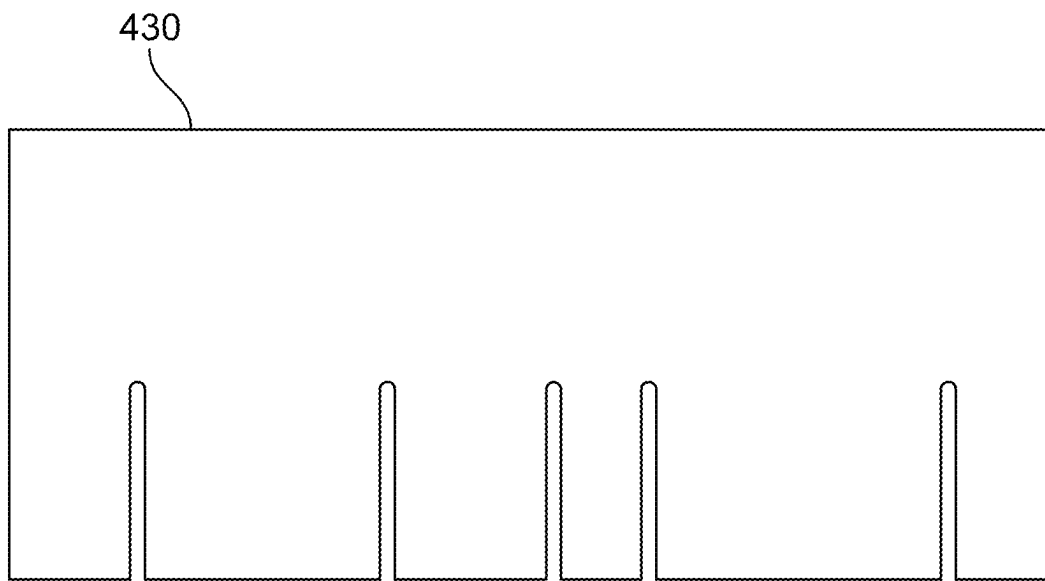


FIG. 4C

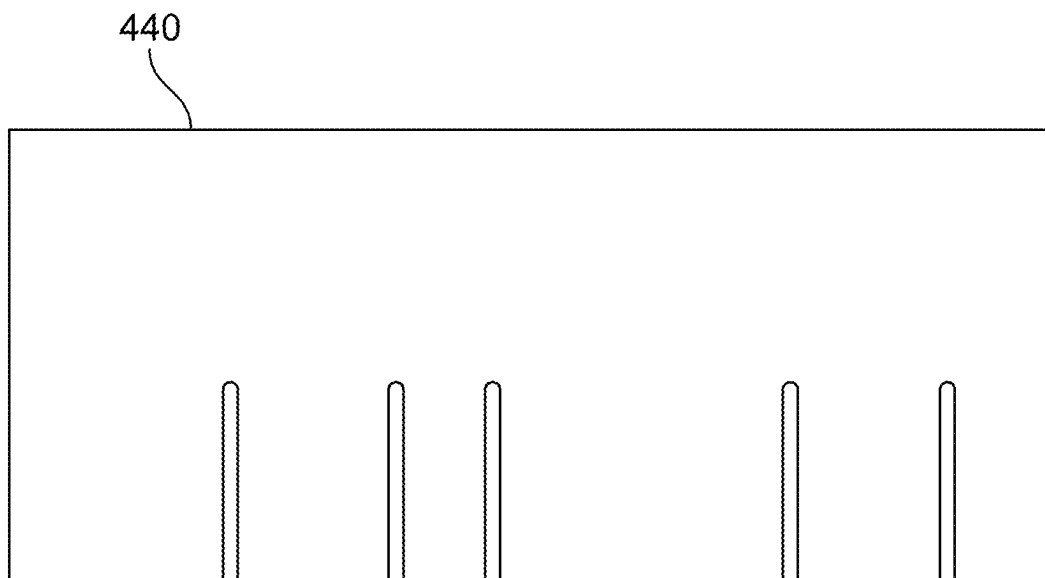


FIG. 4D

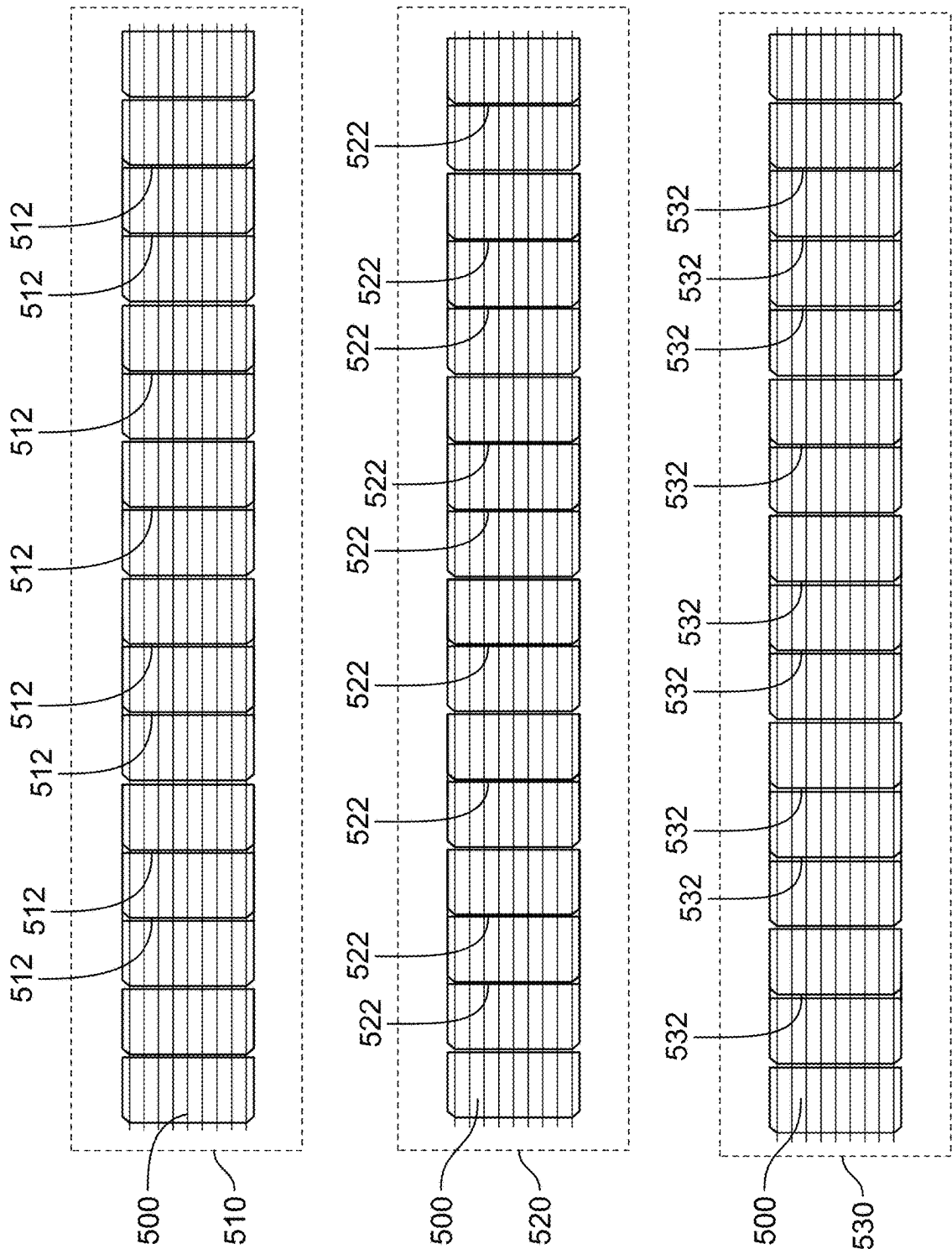


FIG. 5

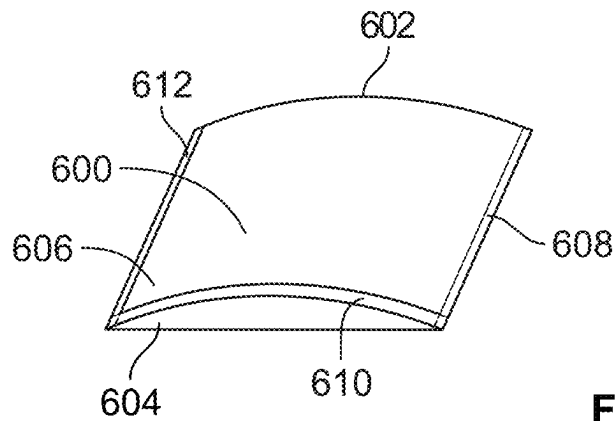


FIG. 6A

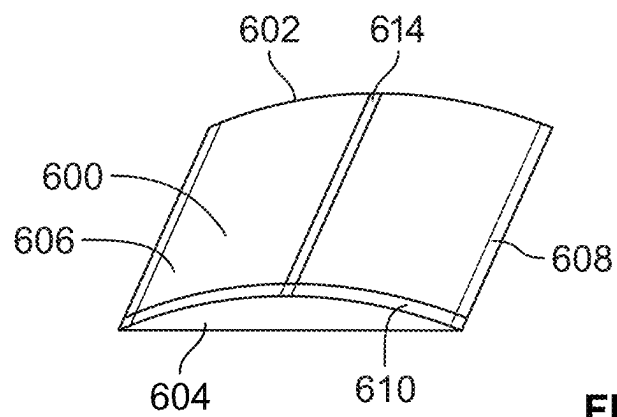


FIG. 6B

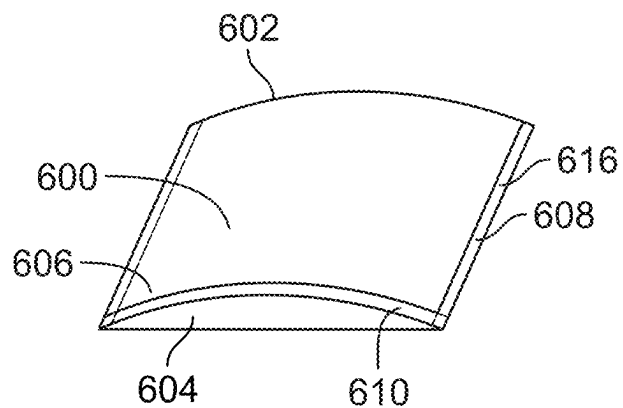


FIG. 6C

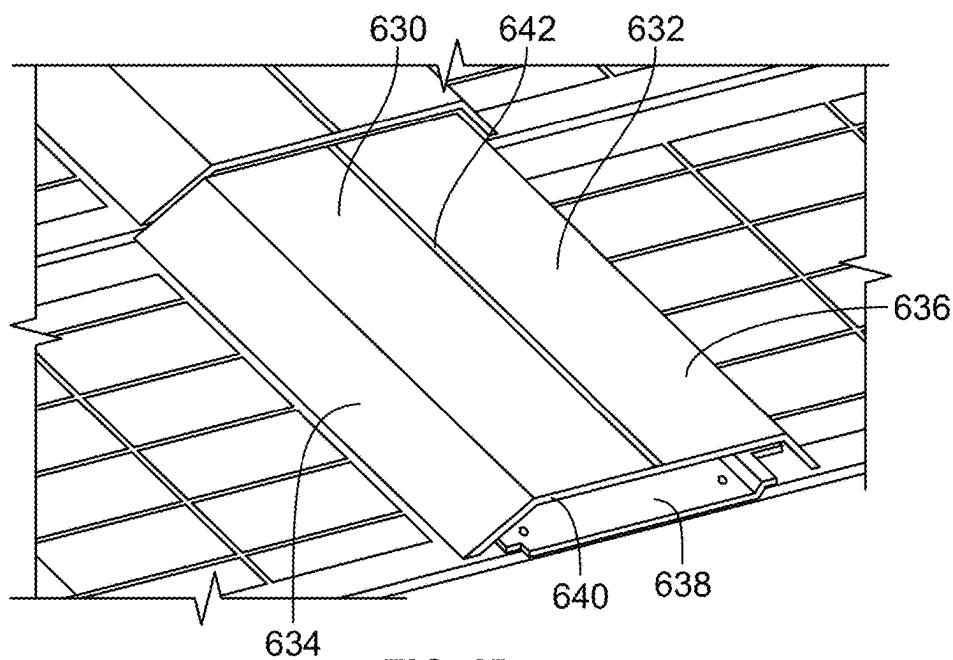


FIG. 6D

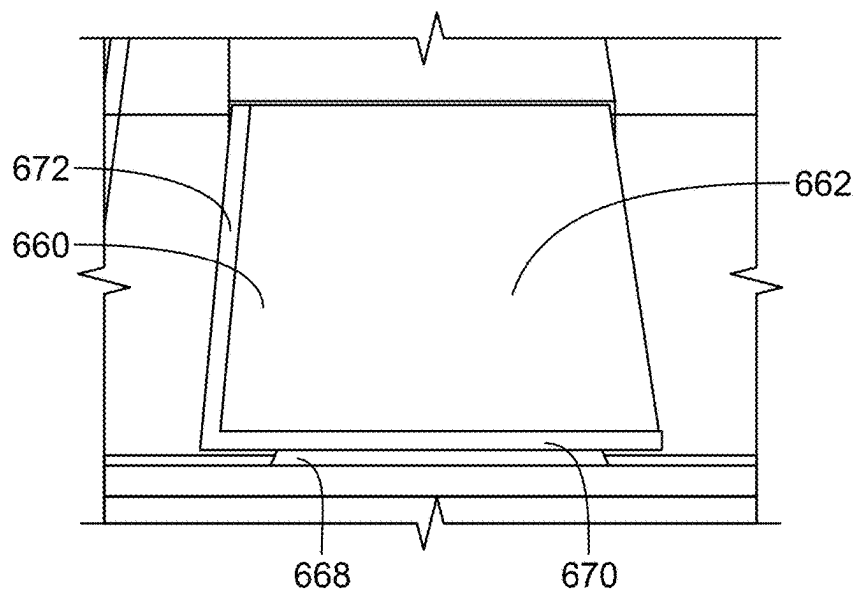


FIG. 6E

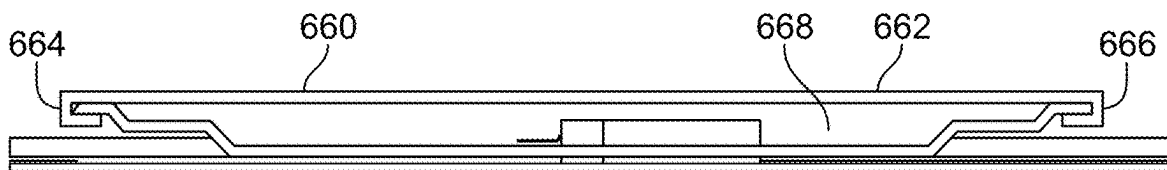


FIG. 6F

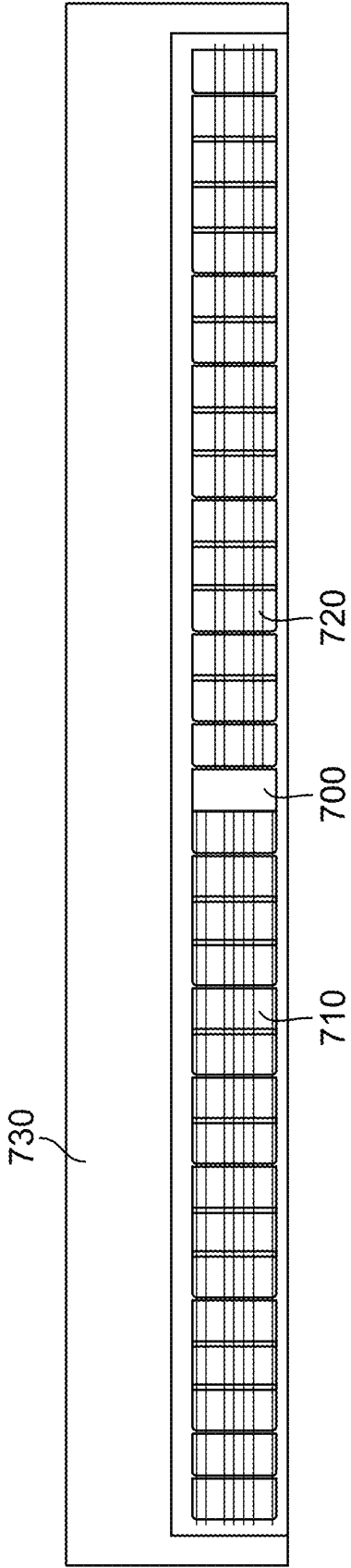


FIG. 7

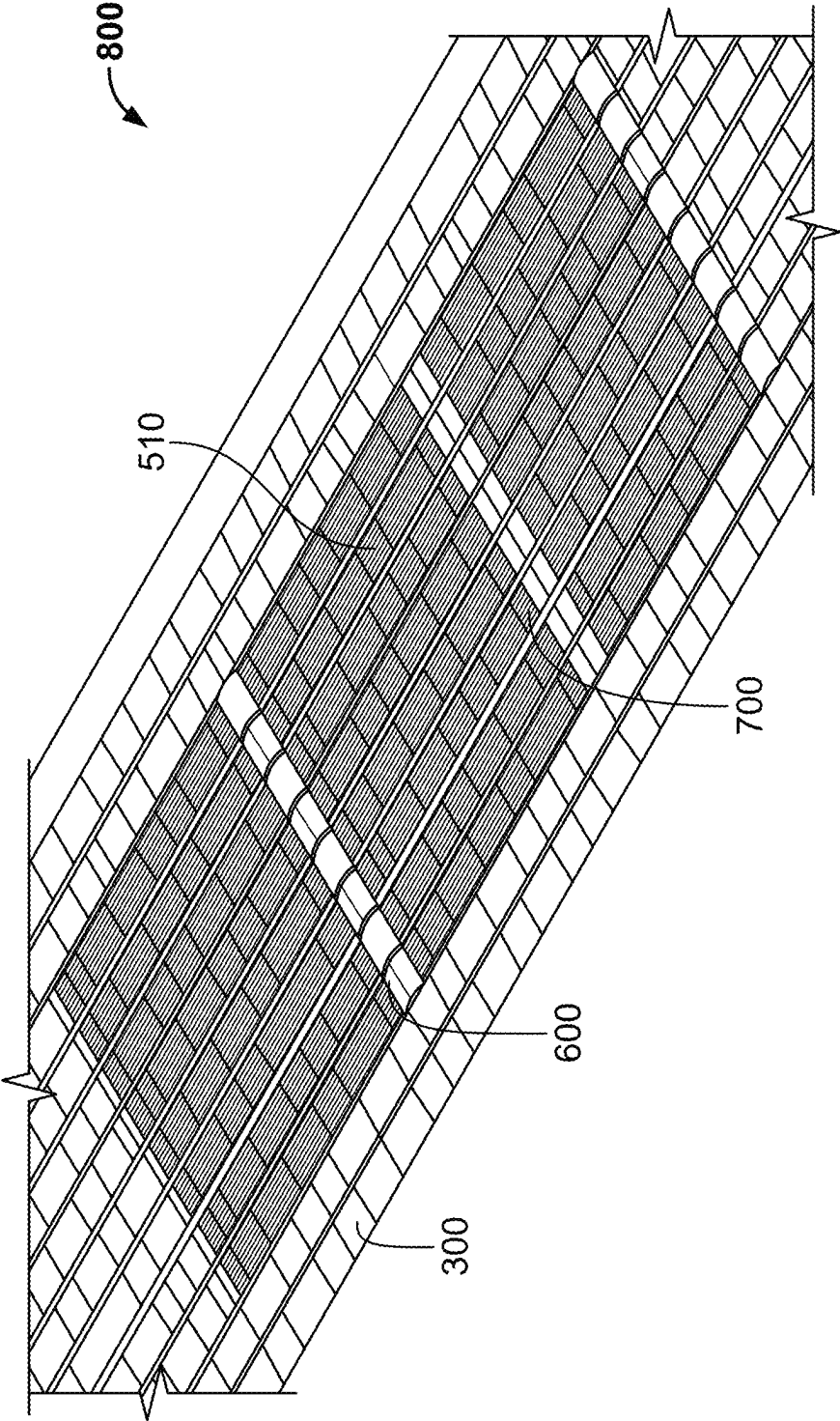


FIG. 8A

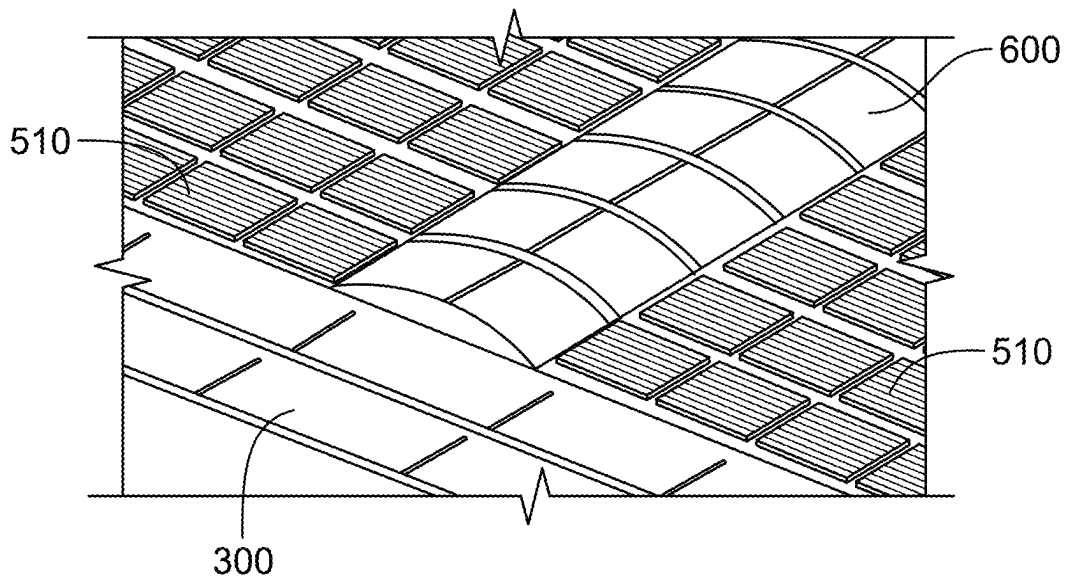


FIG. 8B

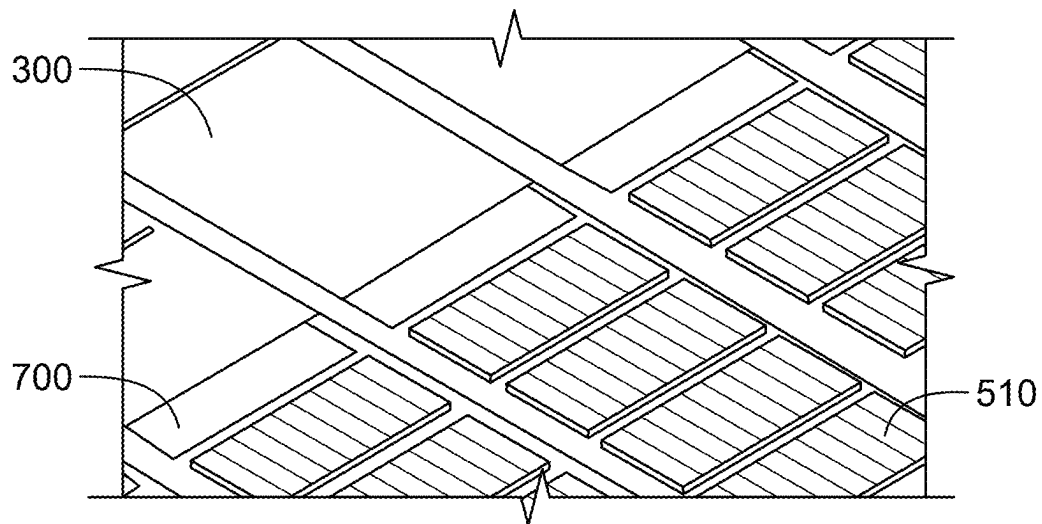


FIG. 8C

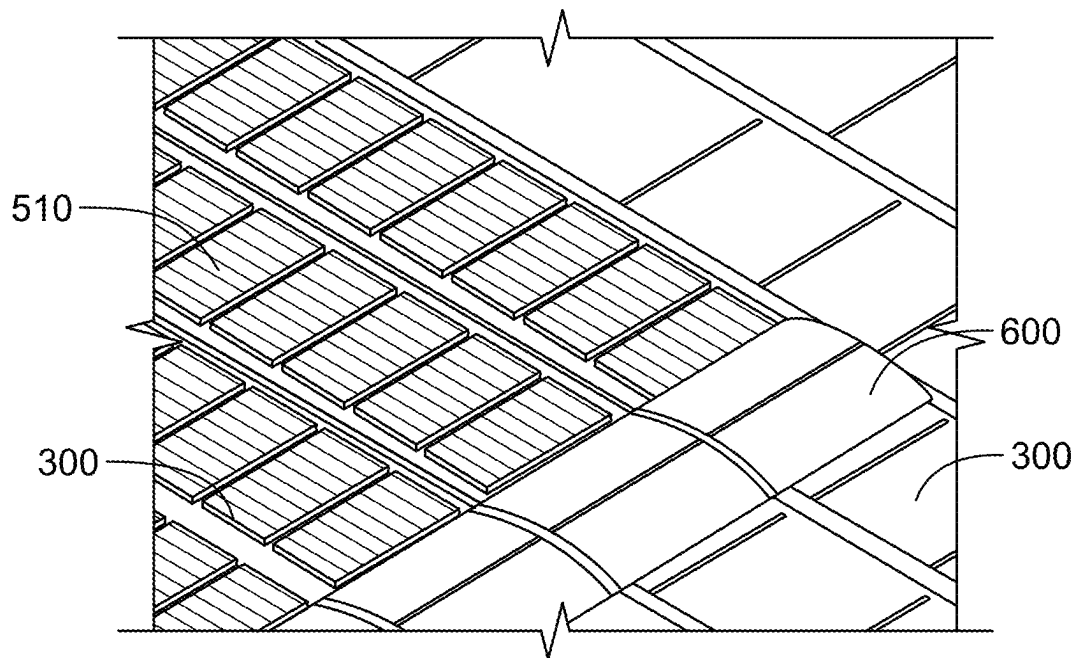


FIG. 8D

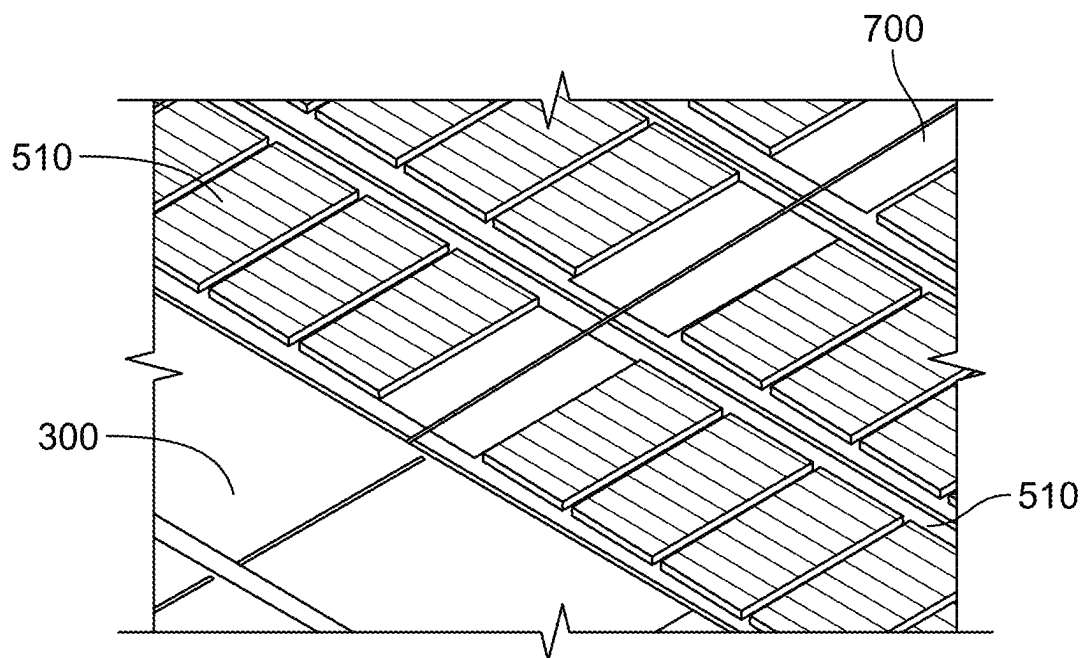


FIG. 8E

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SOLAR ROOFING SYSTEM**CROSS-REFERENCE TO RELATED APPLICATION**

This application is a continuation of U.S. patent application Ser. No. 17/499,516, filed on Oct. 12, 2021, entitled "SOLAR ROOFING SYSTEM", which claims the benefit of U.S. Provisional Patent Application Ser. No. 63/091,017, filed Oct. 13, 2020, entitled "SOLAR ROOFING SYSTEM," the contents of which are incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

The present invention relates to solar roofing systems including roof-integrated photovoltaic modules and roofing shingles. More particularly, the present invention relates to solar roofing systems including roof-integrated photovoltaic modules and roofing shingles having elements with matching widths.

BACKGROUND

Solar modules can be placed on building roofs (e.g., residential roofs) to generate electricity. One obstacle to mass-market adoption of solar roofing is poor aesthetics. Standard rack-mounted photovoltaic ("PV") systems have a very different appearance than traditional roofing materials (e.g., asphalt shingles, wooden shingles, slate shingles, etc.), which can draw unwanted attention. Even low-profile PV systems still receive poor aesthetic feedback from consumers.

Specifically, typical PV module materials and circuit formations include PV elements having a constant width and a grid-like appearance, while typical roofing shingles include elements having irregular viewed widths, causing the PV modules not to visually match the look of standard roofing shingles.

SUMMARY

In some embodiments, a system, comprising a photovoltaic module comprising a plurality of photovoltaic cells, wherein each of the plurality of photovoltaic cells has a photovoltaic cell width; and a roofing shingle having a top surface and a bottom surface, the roofing shingle having an exposure zone at a lower end of the top surface and a headlap zone at an upper end of the top surface, wherein a plurality of slots extends from the lower end toward the headlap zone, wherein the plurality of slots defines a plurality of tooth portions therebetween, wherein a first one of the plurality of tooth portions has a first side that is defined by a first one of the plurality of slots and a second side that is defined by a second one of the plurality of slots that is adjacent to the first one of the plurality of slots, wherein the first one of the plurality of tooth portions has a first width that is measured from the first one of the plurality of slots to the second one of the plurality of slots, wherein the first width is the photovoltaic cell width multiplied by a first positive integer, wherein a second one of the plurality of tooth portions has a first side that is defined by a third one of the plurality of slots and a second side that is defined by a fourth one of the plurality of slots that is adjacent to the third one of the plurality of slots, wherein the second one of the plurality of tooth portions has a second width that is measured from the third one of the plurality of slots to the fourth one of the

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plurality of slots, and wherein the second width is the photovoltaic cell width multiplied by a second positive integer that is different than the first positive integer.

In some embodiments, a third one of the plurality of tooth portions has a first side that is defined by a fifth one of the plurality of slots and a second side that is defined by a sixth one of the plurality of slots that is adjacent to the fifth one of the plurality of slots, and wherein the third one of the plurality of tooth portions has a third width that is measured from the fifth one of the plurality of slots to the sixth one of the plurality of slots, wherein the third width is the photovoltaic cell width multiplied by 0.5 and by a third positive integer that is different than the first positive integer and different than the second positive integer. In some embodiments, each of the first and second positive integers is selected from the group consisting of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, and 20. In some embodiments, the roofing shingle comprises thermoplastic olefin, polyvinyl chloride, or asphalt. In some embodiments, the top surface of the roofing shingle comprises embedded granules.

In some embodiments, a third one of the plurality of tooth portions has a first side that is defined by a fifth one of the plurality of slots and a second side that is defined by one of either (a) the first side, or (b) the second side, and wherein the third one of the plurality of tooth portions has a third width that is measured from the fifth one of the plurality of slots to the one of either the first side or the second side, wherein the third width is the photovoltaic cell width multiplied by 0.5 and by a third positive integer that is different than the first positive integer and different than the second positive integer.

In some embodiments, the system further includes a second roofing shingle having a top surface and a bottom surface, the second roofing shingle having an exposure zone at a lower end of the top surface of the second roofing shingle and a headlap zone at an upper end of the top surface of the second roofing shingle, wherein a plurality of slots extends from the lower end of the second roofing shingle toward the headlap zone of the second roofing shingle, wherein the plurality of slots of the second roofing shingle defines a plurality of tooth portions therebetween, wherein a first one of the plurality of tooth portions of the second roofing shingle has a first side that is defined by a first one of the plurality of slots of the second roofing shingle and a second side that is defined by a second one of the plurality of slots of the second roofing shingle that is adjacent to the first one of the plurality of slots of the second roofing shingle, wherein the first one of the plurality of tooth portions of the second roofing shingle has a third width that is measured from the first one of the plurality of slots of the second roofing shingle to the second one of the plurality of slots of the second roofing shingle, wherein the third width is the photovoltaic cell width multiplied by a third positive integer, wherein a second one of the plurality of tooth portions of the second roofing shingle has a first side that is defined by a third one of the plurality of slots of the second roofing shingle and a second side that is defined by a fourth one of the plurality of slots of the second roofing shingle that is adjacent to the third one of the plurality of slots of the second roofing shingle, wherein the second one of the plurality of tooth portions of the second roofing shingle has a fourth width that is measured from the third one of the plurality of slots of the second roofing shingle to the fourth one of the plurality of slots of the second roofing shingle,

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and wherein the fourth width is the photovoltaic cell width multiplied by a fourth positive integer that is different than the third positive integer.

In some embodiments, an arrangement of the tooth portions of the second roofing shingle is not identical to an arrangement of the tooth portions of the roofing shingle. In some embodiments, an arrangement of the tooth portions of the second roofing shingle is identical to an arrangement of the tooth portions of the roofing shingle. In some embodiments, the third one of the plurality of slots is a same one of the plurality of slots as the second one of the plurality of slots, and wherein the first one of the plurality of tooth portions is adjacent to the second one of the plurality of tooth portions.

In some embodiments, the system further includes a wireway configured to be positioned between the photovoltaic module and a further photovoltaic module that is adjacent to the photovoltaic module, wherein the wireway is configured to enclose at least one electrical cable, wherein a width of the wireway as measured in a horizontal direction between the photovoltaic module and the further photovoltaic module is the photovoltaic cell width multiplied by two, wherein the wireway includes a dark colored portion and a light colored portion, and wherein the light colored portion extends across the wireway in a vertical direction that is perpendicular to the horizontal direction. In some embodiments, the light-colored portion is positioned at an edge of the wireway that is adjacent to the photovoltaic module. In some embodiments, the light-colored portion is positioned halfway intermediate (1) an edge of the wireway that is adjacent to the photovoltaic module and (2) an edge of the wireway that is adjacent to the further photovoltaic module. In some embodiments, the wireway further comprises a further light colored portion extending across a bottom edge of the wireway in the horizontal direction.

In some embodiments, a roofing shingle includes a top surface and a bottom surface, the roofing shingle having an exposure zone at a lower end of the top surface and a headlap zone at an upper end of the top surface, wherein the roofing shingle is configured to be installed on a roof adjacent to a photovoltaic module including a plurality of photovoltaic cells, wherein each of the plurality of photovoltaic cells has a photovoltaic cell width, wherein a plurality of slots extends from the lower end toward the headlap zone, wherein the plurality of slots defines a plurality of tooth portions therebetween, wherein a first one of the plurality of tooth portions has a first side that is defined by a first one of the plurality of slots and a second side that is defined by a second one of the plurality of slots that is adjacent to the first one of the plurality of slots, wherein the first one of the plurality of tooth portions has a first width that is measured from the first one of the plurality of slots to the second one of the plurality of slots, wherein the first width is the photovoltaic cell width multiplied by a first positive integer, wherein a second one of the plurality of tooth portions has a first side that is defined by a third one of the plurality of slots and a second side that is defined by a fourth one of the plurality of slots that is adjacent to the third one of the plurality of slots, wherein the second one of the plurality of tooth portions has a second width that is measured from the third one of the plurality of slots to the fourth one of the plurality of slots, and wherein the second width is the photovoltaic cell width multiplied by a second positive integer that is different than the first positive integer.

In some embodiments, a third one of the plurality of tooth portions has a first side that is defined by a fifth one of the plurality of slots and a second side that is defined by a sixth

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one of the plurality of slots that is adjacent to the fifth one of the plurality of slots, wherein the third one of the plurality of tooth portions has a third width that is measured from the fifth one of the plurality of slots to the sixth one of the plurality of slots, and wherein the third width is the photovoltaic cell width multiplied by a third positive integer that is different than the first positive integer and different than the second positive integer. In some embodiments, each of the first and second positive integers is selected from the group consisting of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, and 20. In some embodiments, the roofing shingle comprises thermoplastic olefin, polyvinyl chloride, or asphalt. In some embodiments, the top surface of the roofing shingle comprises embedded granules. In some embodiments, a third one of the plurality of tooth portions has a first side that is defined by a fifth one of the plurality of slots and a second side that is defined by one of either (a) the first side, or (b) the second side, and wherein the third one of the plurality of tooth portions has a third width that is measured from the fifth one of the plurality of slots to the one of either the first side or the second side, wherein the third width is the photovoltaic cell width multiplied by a third positive integer that is different than the first positive integer and different than the second positive integer.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 shows a perspective view of an exemplary PV module.

FIG. 2A shows a schematic view of elements of a layered structure of an exemplary PV module before lamination.

FIG. 2B shows a schematic view of a layered structure of an exemplary PV module formed by lamination of the elements shown in FIG. 2A.

FIG. 3 shows an exemplary roofing shingle.

FIG. 4A shows a first variant of an exemplary roofing shingle.

FIG. 4B shows a second variant of an exemplary roofing shingle.

FIG. 4C shows a third variant of an exemplary roofing shingle.

FIG. 4D shows a fourth variant of an exemplary roofing shingle.

FIG. 5 shows exemplary embodiments of a PV module.

FIG. 6A shows an exemplary embodiment of a wireway.

FIG. 6B shows an exemplary embodiment of a wireway.

FIG. 6C shows an exemplary embodiment of a wireway.

FIGS. 6D through 6F show another exemplary embodiment of a wireway.

FIG. 7 shows an exemplary embodiment of a masking element.

FIG. 8A shows an exemplary embodiment of a roofing system.

FIG. 8B shows a magnified view of a first portion of the roofing system of FIG. 8A.

FIG. 8C shows a magnified view of a second portion of the roofing system of FIG. 8A.

FIG. 8D shows a magnified view of a third portion of the roofing system of FIG. 8A.

FIG. 8E shows a magnified view of a fourth portion of the roofing system of FIG. 8A.

DETAILED DESCRIPTION

The present invention will be further explained with reference to the attached drawings, wherein like structures are referred to by like numerals throughout the several

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views. The drawings shown are not necessarily to scale, with emphasis instead generally being placed upon illustrating the principles of the present invention. Further, some features may be exaggerated to show details of particular components.

The figures constitute a part of this specification and include illustrative embodiments of the present invention and illustrate various objects and features thereof. Further, the figures are not necessarily to scale, some features may be exaggerated to show details of particular components. In addition, any measurements, specifications and the like shown in the figures are intended to be illustrative, and not restrictive. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the present invention.

Among those benefits and improvements that have been disclosed, other objects and advantages of this invention will become apparent from the following description taken in conjunction with the accompanying figures. Detailed embodiments of the present invention are disclosed herein; however, it is to be understood that the disclosed embodiments are merely illustrative of the invention that may be embodied in various forms. In addition, each of the examples given in connection with the various embodiments of the invention which are intended to be illustrative, and not restrictive.

Throughout the specification and claims, the following terms take the meanings explicitly associated herein, unless the context clearly dictates otherwise. The phrases “in one embodiment” and “in some embodiments” as used herein do not necessarily refer to the same embodiment(s), though they may. Furthermore, the phrases “in another embodiment” and “in some other embodiments” as used herein do not necessarily refer to a different embodiment, although they may. Thus, as described below, various embodiments of the invention may be readily combined, without departing from the scope or spirit of the invention.

The term “based on” is not exclusive and allows for being based on additional factors not described, unless the context clearly dictates otherwise. In addition, throughout the specification, the meaning of “a,” “an,” and “the” include plural references. The meaning of “in” includes “in” and “on.”

The exemplary embodiments relate to a roofing system having elements (e.g., roofing shingles, PV modules, wireways, and masking elements) having visual elements the width of which is harmonized around a constant base width. In some embodiments, the constant base width is a cell pitch. In some embodiments, the cell pitch is defined as the width of a PV element plus the width of a space that exists between two adjacent PV elements. In some embodiments, such harmonization provides a consistent and blended aesthetic appearance across such a roofing system, as will be discussed in further detail hereinafter.

In some embodiments, a solar roofing system includes at least one PV module and at least one roofing shingle. In some embodiments, each of the at least one PV modules includes a plurality of PV cells. In some embodiments, each of the PV cells has a PV cell width that is the same for all of the PV cells. In some embodiments, the cell width is a “half-cut” width, e.g., the width of a premanufactured PV cell that has been cut in half.

FIG. 1 shows an exemplary PV module 100. The exemplary PV module 100 includes a headlap region 110 and a PV region 120. In some embodiments, the headlap region 110 comprises thermoplastic olefin (“TPO”), polyvinyl chloride (“PVC”), or asphalt. In some embodiments, the headlap

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region 110 includes embedded granules. In some embodiments, the headlap region 110 defines a nailing line 112 extending across the headlap region 110. In some embodiments, the nailing line 112 extends across the headlap region 110 approximately midway between the end of the headlap region 110 that borders the PV region and the opposite end of the headlap region 110. In some embodiments, the nailing line 112 defines an area of the headlap region 110 through which mechanical fasteners (e.g., nails, screws, etc.) can be driven to secure the PV module 100 to a roof deck in the standard manner.

In some embodiments, the PV region 120 includes a plurality of PV portions 122. In some embodiments, each of the PV portions 122 includes a layered structure that is typical of a laminate PV module, as discussed below with reference to FIGS. 2A and 2B. In some embodiments, the PV region 120 includes grooves 124 separating adjacent ones of the PV portions. In some embodiments, each of the PV portions 122 is separately formed from others of the PV portions 122, and the grooves 124 are formed by spaces between adjacent ones of the PV portions 122. In some embodiments, the PV portions 122 forming the PV region 120 are integrally formed with one another (e.g., form a single layered structure) and the grooves 124 are formed in a superstrate layer thereof. In some embodiments, the grooves 124 between adjacent ones of the PV portions 122 provide the appearance of discrete portions similar to those of conventional shingles. In some embodiments, the PV region 120 is formed atop material of the headlap region 110 (e.g., the substrate of the PV region 120 is deposited on the material of the headlap region 110). In some embodiments, the PV region 120 and the headlap region 110 join one another end-to-end.

In some embodiments, the PV module 100 includes a junction box 130. In some embodiments, the junction box 130 is positioned at an end of the headlap region 110 that is opposite the PV region 120. In some embodiments, the junction box 130 is positioned at a center of an end of the headlap region 110 that is opposite the PV region 120. In some embodiments, the junction box 130 is electrically connected to the PV region 120 by electrical connectors (e.g., wires) that traverse (e.g., pass under or through) the headlap region 110.

In some embodiments, such as the PV module 100 discussed above with reference to FIG. 1, an exemplary PV module includes a layered structure. FIGS. 2A and 2B show an exemplary embodiment of a layered structure 200 that, in some embodiments, forms part of an exemplary PV module.

FIG. 2A shows an exploded view of the layers of the layered structure 200 prior to lamination to form the layered structure 200. FIG. 2B shows the layered structure following lamination. It will be apparent to those of skill in the art that FIGS. 2A and 2B present schematic views of the layered structure 200 and are not intended to provide a to-scale representation.

Referring now to FIG. 2A, in some embodiments, the layered structure 200 includes a superstrate layer 210 that forms an upper surface of the layered structure 200 and of the PV module 100 (i.e., the surface that, when the PV module 100 is installed on a roof, faces away from the roof and toward the sun). In some embodiments, the superstrate layer 210 has an upper surface 212 (i.e., the side of the superstrate layer 210 that faces toward the sun when installed as described above) and a lower surface 214 opposite the upper surface 212. In some embodiments, the superstrate layer 210 is optically transparent (e.g., it has a solar weighted transmittance of 80% or greater). In some

embodiments, the superstrate provides electrical insulation and moisture resistance. In some embodiments, the superstrate layer **210** comprises a glass material, such as low-iron solar glass. In some embodiments, the superstrate layer **210** comprises a polymeric material such as ethylene tetrafluoroethylene (“ETFE”), polyethylene terephthalate (“PET”), or an acrylic such as polymethyl methacrylate (“PMMA”). In some embodiments, the superstrate layer **210** has a thickness of from 50 microns to 250 microns. In some embodiments, the superstrate layer **210** has a thickness of from 50 microns to 200 microns. In some embodiments, the superstrate layer **210** has a thickness of from 50 microns to 150 microns. In some embodiments, the superstrate layer **210** has a thickness of from 50 microns to 100 microns. In some embodiments, the superstrate layer **210** has a thickness of from 100 microns to 250 microns. In some embodiments, the superstrate layer **210** has a thickness of from 100 microns to 200 microns. In some embodiments, the superstrate layer **210** has a thickness of from 100 microns to 150 microns. In some embodiments, the superstrate layer **210** has a thickness of from 150 microns to 250 microns. In some embodiments, the superstrate layer **210** has a thickness of from 150 microns to 200 microns. In some embodiments, the superstrate layer **210** has a thickness of from 200 microns to 250 microns.

In some embodiments, the superstrate layer **210** has a thickness of from 200 microns to 500 microns. In some embodiments, the superstrate layer **210** has a thickness of from 200 microns to 450 microns. In some embodiments, the superstrate layer **210** has a thickness of from 200 microns to 400 microns. In some embodiments, the superstrate layer **210** has a thickness of from 200 microns to 350 microns. In some embodiments, the superstrate layer **210** has a thickness of from 200 microns to 300 microns. In some embodiments, the superstrate layer **210** has a thickness of from 250 microns to 500 microns. In some embodiments, the superstrate layer **210** has a thickness of from 250 microns to 450 microns. In some embodiments, the superstrate layer **210** has a thickness of from 250 microns to 400 microns. In some embodiments, the superstrate layer **210** has a thickness of from 250 microns to 350 microns. In some embodiments, the superstrate layer **210** has a thickness of from 250 microns to 300 microns. In some embodiments, the superstrate layer **210** has a thickness of from 300 microns to 500 microns. In some embodiments, the superstrate layer **210** has a thickness of from 300 microns to 450 microns. In some embodiments, the superstrate layer **210** has a thickness of from 300 microns to 400 microns. In some embodiments, the superstrate layer **210** has a thickness of from 300 microns to 350 microns. In some embodiments, the superstrate layer **210** has a thickness of from 350 microns to 500 microns. In some embodiments, the superstrate layer **210** has a thickness of from 350 microns to 450 microns. In some embodiments, the superstrate layer **210** has a thickness of from 350 microns to 400 microns. In some embodiments, the superstrate layer **210** has a thickness of from 400 microns to 500 microns. In some embodiments, the superstrate layer **210** has a thickness of from 400 microns to 450 microns. In some embodiments, the superstrate layer **210** has a thickness of from 450 microns to 500 microns. In some embodiments, the superstrate layer **210** has a thickness of from 325 microns to 375 microns. In some embodiments, the superstrate layer **210** has a thickness of about 300 microns. In some embodiments, the superstrate layer **210** has a thickness of 300 microns.

In some embodiments, the superstrate layer **210** has a thickness of from 1.6 millimeters to 4 millimeters. In some embodiments, the superstrate layer **210** has a thickness of from 1.6 millimeters to 3.2 millimeters. In some embodiments, the superstrate layer **210** has a thickness of from 1.6 millimeters to 2.4 millimeters. In some embodiments, the superstrate layer **210** has a thickness of from 2.4 millimeters to 4 millimeters. In some embodiments, the superstrate layer **210** has a thickness of from 2.4 millimeters to 3.2 millimeters. In some embodiments, the superstrate layer **210** has a thickness of from 3.2 millimeters to 4 millimeters. In some embodiments, the superstrate layer **210** has a thickness of from 2.8 millimeters to 3.6 millimeters. In some embodiments, the superstrate layer **210** has a thickness of from 3 millimeters to 3.4 millimeters. In some embodiments, the superstrate layer **210** has a thickness of from 3.1 millimeters to 3.3 millimeters. In some embodiments, the superstrate layer **210** has a thickness about 3.2 millimeters. In some embodiments, the superstrate layer **210** has a thickness of 3.2 millimeters.

Continuing to refer to FIG. 2A, in some embodiments, the layered structure **200** includes an upper encapsulant layer **220**. In some embodiments, the upper encapsulant layer **220** has an upper surface **222** and a lower surface **224** opposite the upper surface **222**. In some embodiments, the upper surface **222** of the upper encapsulant layer **220** contacts the lower surface **214** of the superstrate layer **210**. In some embodiments, the upper encapsulant layer **220** is optically transparent (e.g., it has a solar weighted transmittance of 80% or greater). In some embodiments, the upper encapsulant layer provides electrical insulation. In some embodiments, the upper encapsulant layer **220** comprises an encapsulating material such as ethylene-co-vinyl acetate (“EVA”), polydimethyl siloxane (“PDMS”), a polyolefin elastomer (“POE”), polyvinyl butyral (“PVB”), polyurethane epoxy, silicone, or an ionomer such as the series of ionomer-based encapsulants commercialized by DuPont de Nemours, Inc. under the trade name PV5400. In some embodiments, the thickness of the upper encapsulant layer **220** varies across the layered structure **200**, as will be discussed in greater detail hereinafter.

Continuing to refer to FIG. 2A, in some embodiments, the layered structure **200** includes a PV layer **230** having an upper surface **232** and a lower surface **234** opposite the upper surface **232**. In some embodiments, the upper surface **232** of the PV layer **230** contacts the lower surface **224** of the upper encapsulant layer **220**. In some embodiments, the PV layer **230** includes at least one PV element **236** (e.g., at least one PV cell having a constant PV cell width **237** as described above). In some embodiments, the PV layer **230** includes an array of PV elements **236**. In some embodiments in which the PV layer **230** includes a plurality of the PV element **236**, the PV elements **236** are electrically interconnected with one another. In some embodiments, the PV layer **230** includes an array of interconnected PV elements **236**. In some embodiments, gaps are formed between adjacent ones of the PV elements **236**. In some embodiments, the gaps are significantly smaller than the PV elements **236**; for example, in some embodiments, a width of each of the PV elements **236** is 160 millimeters and the gaps are from 2 millimeters to 5 millimeters in size. In some embodiments, the PV layer **230** also includes other active and/or passive electronic components.

Continuing to refer to FIG. 2A, in some embodiments, the layered structure **200** includes a lower encapsulant layer **240** having an upper surface **242** and a lower surface **244** opposite the upper surface **242**. In some embodiments, the

upper surface 242 of the lower encapsulant layer 240 contacts the lower surface 234 of the PV layer 230. In some embodiments, the lower encapsulant layer 240 provides electrical insulation. In some embodiments, the lower encapsulant layer 240 is optically transparent. In some embodiments, the lower encapsulant layer 240 is not optically transparent. In some embodiments, the thickness of the lower encapsulant layer 240 is in the range of 100 to 1000 microns. In some embodiments, the thickness of the lower encapsulant layer 240 is sufficiently large (e.g., greater than 100 microns) so as to prevent delamination between the PV layer 230 and the substrate 250. In some embodiments, the thickness of the lower encapsulant layer 240 is consistent across the entirety of the layered structure 200. In some embodiments, the lower encapsulant layer 240 comprises an encapsulating material such as ethylene-co-vinyl acetate ("EVA"), polydimethyl siloxane ("PDMS"), a polyolefin elastomer ("POE"), polyvinyl butyral ("PVB"), polyurethane epoxy, silicone, or an ionomer such as the series of ionomer-based encapsulants commercialized by DuPont de Nemours, Inc. under the trade name PV5400. In some embodiments, the lower encapsulant layer 240 comprises the same encapsulating material as the upper encapsulant layer 220.

Continuing to refer to FIG. 2A, in some embodiments, the layered structure 200 includes a substrate 250 having an upper surface 252 and a lower surface 254 opposite the lower surface 252. In some embodiments, the upper surface 252 of the substrate 250 contacts the lower surface 244 of the lower encapsulant layer 240. In some embodiments, the lower surface 254 of the substrate 250 forms the lower surface 204 of the layered structure 200. In some embodiments, the substrate 250 provides electrical insulation and moisture resistance. In some embodiments, the substrate 250 is optically transparent. In some embodiments, the substrate 250 is not optically transparent. In some embodiments, the substrate 250 comprises a glass material. In some embodiments, the substrate 250 comprises a polymeric material such as ETFE, PET, an acrylic such as PMMA, polypropylene, polyvinyl chloride ("PVC"), or a glass-reinforced or fiber-reinforced composite such as a material meeting the National Electrical Manufacturers Association ("NEMA") grades FR-4 or G-10. In some embodiments, the substrate 250 has a thickness in the range of 200 microns to ¼ inch. In some embodiments, the substrate 250 is sufficiently rigid to provide mechanical stiffening to the PV module 100.

Referring now to FIG. 2B, the layered structure 200 is shown following lamination. In some embodiments, during the lamination process, the encapsulating material of the upper encapsulant layer 220 and the encapsulating material of the lower encapsulant layer 240 are melted and flow within the gaps between adjacent ones of the PV elements 236 shown in FIG. 2A, thereby encapsulating (e.g., surrounding on all sides) each of the PV elements 236 with encapsulating material. In some embodiments, as a result of this process, the PV layer 230 includes encapsulant portions 238 located between adjacent ones of the PV elements 236, and providing continuity between the encapsulating material of the upper encapsulant layer 220 and the encapsulating material of the lower encapsulant layer 240. In some embodiments, the resulting region of the layered structure 200 (e.g., the upper encapsulant layer 220, the PV layer 230, and the lower encapsulant layer 240) resembles a single block of encapsulant material with the PV elements positioned therein.

FIG. 3 shows an exemplary roofing shingle 300. In some embodiments, the roofing shingle 300 comprises thermo-

plastic olefin ("TPO"), polyvinyl chloride ("PVC"), or asphalt. In some embodiments, roofing shingle 300 includes embedded granules. In some embodiments, the roofing shingle 300 includes a top end 302, a bottom end 304, a first side 306, and a second side. In some embodiments, the roofing shingle 300 includes a headlap region 310 adjacent the top end 302 and a tooth region 320 adjacent the bottom end 304. In some embodiments, the headlap region 310 defines a nailing line 312 extending across the headlap region 310. In some embodiments, the nailing line 312 extends across the headlap region 310 approximately midway between the end of the headlap region 310 that borders the tooth region 320 and the opposite end of the headlap region 110. In some embodiments, the nailing line 312 defines an area of the headlap region 310 through which mechanical fasteners (e.g., nails, screws, etc.) can be driven to secure the roofing shingle 300 to a roof deck in the standard manner.

In some embodiments, the tooth region 320 includes a plurality of slots 322 that are spaced apart along the width of the tooth region 320 from the first side 306 to the second side, and extend from the bottom end 304 toward the top end 302. In some embodiments, each of the slots 322 has a width that is the same as, or is similar to, the gaps between adjacent ones of the PV elements 236 in the PV module 100. In some embodiments, the width of each of the slots 322 is ½ inch. In some embodiments, the width of each of the slots 322 is 6 millimeters. In some embodiments, the first side 306, the slots 322, and the second side define a plurality of tooth portions 324, 326, 328, 330, 332, 334 therebetween. The exemplary roofing shingle 300 shown in FIG. 3 includes six (6) of the tooth portions 324, 326, 328, 330, 332, 334, but it will be apparent to those of skill in the art that various embodiments of the roofing shingle 300 may have any other number of tooth portions. In some embodiments, the width of the tooth portions 324, 326, 328, 330, 332, 334 is variable, e.g., for a given instance of the roofing shingle 300, a first one of the tooth portions (e.g., tooth portion 324) has a first width, and a second one of the tooth portions (e.g., tooth portion 326) has a second width that is different than the first width. In some embodiments, each of the tooth portions 324, 326, 328, 330, 332, 334 has a width that is an integer multiple of the PV cell width 237 (e.g., is equal to the PV cell width 237, or is two times, or three times, or four times, or five times, or six times, or seven times, or eight times, or nine times, or ten times, or eleven times, or twelve times, or thirteen times, or fourteen times, or fifteen times, or sixteen times, or seventeen times, or eighteen times, or nineteen times, or twenty times the PV cell width 237).

In some embodiments, different ones of the exemplary roofing shingle 300 have differently sized and differently arranged tooth portions. For example, in some embodiments, a manufacturer of the roofing shingle 300 may manufacture different versions of the roofing shingle 300 so as to provide a roofing system including a plurality of the roofing shingle 300 that differ from one another so as to provide a non-uniform appearance to the roofing system. FIGS. 4A-4D show different variants 410, 420, 430, and 440 of the exemplary roofing shingle 300. In some embodiments, each of the variants 410, 420, 430, and 440 has a plurality of tooth portions that vary in width among the different tooth portions of any given one of the variants 410, 420, 430, and 440, and the widths of all tooth portions are integer multiples of the PV cell width 237. In some embodiments, each of the variants 410, 420, 430, and 440 has tooth portions that differ in arrangement as compared to those of the others of the variants 410, 420, 430, and 440. In some embodiments, a

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manufacturer of the roofing shingle **300** may manufacture a suitable number of different variants so as to impart a random appearance to a roofing system incorporating such variants. For example, in some embodiments, a roofing system includes four of the variants **410**, **420**, **430**, and **440**, as shown in FIGS. **4A-4D**, but it will be apparent to those of skill in the art that this is only exemplary, and that any other number of variants may be produced.

In some embodiments, the roofing shingle **300** is composed of a single layer. In some embodiments, the roofing shingle **300** is composed of multiple layers. In some embodiments, the roofing shingle **300** is laminated.

As discussed above, in some embodiments, the assembled PV module **100** includes a plurality of PV elements **236** that are spaced apart by a quantity of encapsulant portions **238** that are positioned between the PV elements **236** as part of the lamination process. In some embodiments, the space between the PV elements **236** that is formed in this manner is referred to as a “cell gap”. In some embodiments, due to uniform sizing and spacing of the PV elements **236**, the PV elements **236** and the cell gaps therebetween provide a uniform, grid-like appearance. In some embodiments, to provide a non-uniform appearance, the cell gap is selectively revealed or hidden.

FIG. **5** shows PV elements **500** forming PV modules **510**, **520**, **530**. For clarity, the PV elements **500** are substantially the only elements of the PV modules **510**, **520**, **530** shown in FIG. **5**, but it will be apparent to those of skill in the art that the PV modules **510**, **520**, **530** will include other elements as described above. Also for clarity, only one of the PV elements **500** forming each of the PV modules **510**, **520**, **530** is specifically called out in FIG. **5**, but it will be apparent to those of skill in the art that discussion of the PV elements **500** may refer to any of the PV elements **500** forming the PV modules **510**, **520**, **530**, whether or not specifically identified in FIG. **5**.

In some embodiments, as shown in FIG. **5**, each of the PV modules **510**, **520**, **530** includes sixteen (16) of the PV elements **500** that are “half-cut” cells having a width of about 90 mm and a cell gap of about 4 mm between adjacent ones of the PV elements, thereby to produce the PV modules **510**, **520**, **530** that are 60 inches wide, but it will be apparent to those of skill in the art that these dimensions are only exemplary. In some embodiments, the PV modules **510**, **520**, **530** include portions **512**, **522**, **532**, respectively, of a color-contrasting material positioned behind the PV elements **500** (e.g., positioned in or on the PV modules **510**, **520**, **530** so as to be positioned between the PV elements and a roof deck to which the PV modules **510**, **520**, **530** are installed). In some embodiments, the color-contrasting material comprises a patterned backsheet. In some embodiments, the color-contrasting material comprises a cloaking tape. As shown in FIG. **5**, the inclusion of the portions **512**, **522**, **532** at different locations within the PV modules **510**, **520**, **530** provides a non-uniform appearance to the PV modules **510**, **520**, **530**, despite each of the PV modules **510**, **520**, **530** having the same arrangement of the PV elements. Though FIG. **5** shows three of the PV modules **510**, **520**, **530** having different arrangements of the portions **512**, **522**, **532**, it will be apparent to those of skill in the art that any number of different arrangements are possible without departing from the general concept embodied by the PV modules **510**, **520**, **530**. Throughout this disclosure, the PV module **510** is indicated when it is desired to reference a PV module having color-contrasting material applied to the boundaries between some of the PV elements **500** so as to provide a non-uniform appearance as described above, but such reference to the PV

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module **510** is intended to refer to any of the PV modules **510**, **520**, **530**, or any other PV module having such features.

In some embodiments, an exemplary solar roofing system includes wireways that are positioned between PV modules and are configured to enclose electrical cables that connect to the PV modules. In some embodiments, to facilitate providing an aesthetic appearance that is consistent both for the PV modules and the roofing shingles that form a solar roofing system, an exemplary solar roofing system includes wireways having a width that is matched to a width of the PV elements within the PV modules. FIGS. **6A**, **6B**, and **6C** show embodiments of an exemplary wireway **600** having an upper end **602**, a lower end **604**, a first side **606**, and a second side **608**. In some embodiments, the exemplary wireway **600** is configured to be installed on a roof deck such that the lower end **604** is at a lower elevation than is the upper end **602**. In some embodiments, the exemplary wireway **600** has a width as measured in a horizontal direction from the first side **606** to the second side **608** that is equal to an integer multiple of the PV cell width **237** for PV elements **236** that are used in the same solar roofing system as the exemplary wireway **600**. In some embodiments, the width of the wireway **600** is two times the PV cell width **237**. In some embodiments, the wireway **600** is rounded (e.g., so as to be concave on the side that faces the roof deck) to provide space to accommodate electrical cables and to soften the appearance of the wireway **600**. In some embodiments, the top surface of the wireway **600** (e.g., the side that faces away from the roof deck) includes at least one dark-colored portion and at least one light-colored portion.

In some embodiments, the at least one light-colored portion includes a horizontal light-colored portion **610** extending across the wireway **600** from the first side **606** to the second side **608** adjacent the lower end **604**. In some embodiments, a height of the horizontal light-colored portion **610** (e.g., as measured in a vertical direction from the lower end **604** toward the upper end **602**) is equal to a creepage distance. As used herein, the creepage distance is the shortest distance along the surface of the insulating material between two conductive live parts or between conductive live parts and accessible part. For example, in embodiments detailed herein, the creepage distance is the shortest distance along the surface of an insulative portion of the PV module **510** between two conductive or accessible portions of the PV module **510**. In some embodiments, the creepage distance results in the appearance of a light-colored region along the long edges of the PV module **510**. Consequently, in some embodiments, the sizing of the horizontal light-colored portion **610** as equal to the creepage distance provides continuity of visual appearance between the wireway **600** and PV modules **510** that are adjacent thereto. In some embodiments, the height of the horizontal light-colored portion **610** is in a range of from 10 millimeters to 30 millimeters. In some embodiments, the height of the horizontal light-colored portion **610** is in a range of from 12.5 millimeters to 27.5 millimeters. In some embodiments, the height of the horizontal light-colored portion **610** is in a range of from 15 millimeters to 25 millimeters. In some embodiments, the height of the horizontal light-colored portion **610** is in a range of from 17.5 millimeters to 22.5 millimeters. In some embodiments, the height of the horizontal light-colored portion **610** is about 20 millimeters. In some embodiments, the height of the horizontal light-colored portion **610** is 20 millimeters.

In some embodiments, the at least one light-colored portion includes a vertical light-colored portion extending along the wireway **600** from the upper end **602** toward the

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lower end **604**. In some embodiments, as shown in FIG. 6A, the vertical light-colored portion **612** extends along the wireway **600** and along the first side **606**. In some embodiments, as shown in FIG. 6B, the vertical light-colored portion **614** extends along the wireway **600** and intermediate the first and second sides **606**, **608**. In some embodiments, as shown in FIG. 6C, the vertical light-colored portion **616** extends along the wireway **600** and along the second side **608**. In some embodiments, the width of the vertical light-colored portion **612**, **614**, **616** is about equal to (e.g., within plus or minus 25%) the width of a cell gap, as described above with reference to FIG. 5. In some embodiments, the width of the vertical light-colored portion **612**, **614**, **616** is equal to the width of a cell gap. In some embodiments, the width of the vertical light-colored portion **612**, **614**, **616** is 6 mm. In some embodiments, the width of the vertical light-colored portion **612**, **614**, **616** is about 6 mm. In some embodiments, the width of the vertical light-colored portion **612**, **614**, **616** is from 5 mm to 7 mm. In some embodiments, the width of the vertical light-colored portion **612**, **614**, **616** is from 4 mm to 8 mm. In some embodiments, the width of the vertical light-colored portion **612**, **614**, **616** is from 3 mm to 9 mm. In some embodiments, the width of the vertical light-colored portion **612**, **614**, **616** is from 3% of a width of the wireway **600** to 10% of the width of the wireway **600**. In some embodiments, the vertical light-colored portion **612** or **616** either provides a dark-colored region having a width that is twice the PV cell width **237**. In some embodiments, the vertical light-colored portion **614** provides two dark-colored regions, each of which has a width that is equal to the PV cell width **237**. Accordingly, in some embodiments, vertical light-colored portions **612**, **614**, **616** cause the wireway to have an appearance that is consistent with those of the roofing shingle **300** and the PV module **510**, thereby improving the consistency, visual flow, and aesthetic appearance of a roofing system including the wireway **600**.

In some embodiments, an exemplary wireway **630**, as shown in FIG. 6D positioned between adjacent PV modules, is substantially similar to the wireway **600** described above with reference to FIGS. 6A-6C, other than as described hereinafter. In some embodiments, the wireway **630** has a flat top surface **632** and angled sides **634**, **636**. In some embodiments, the flat top surface **632** and angled sides **634**, **636** of the wireway **630** define a channel **638** within the wireway **630** to provide space to accommodate electrical cables. In some embodiments, the top surface of the wireway **630** (e.g., the side that faces away from the roof deck) includes at least one dark-colored portion and at least one light-colored portion. In some embodiments, the wireway **630** includes a horizontal light-colored portion **640** that is substantially similar to the horizontal light-colored portion **610** of the wireway **600** described above. In some embodiments, the wireway **630** includes a vertical light-colored portion **642**. The vertical light-colored portion **642** shown in FIG. 6D extends across the top surface **632** of the wireway **630** intermediate the sides **634**, **636**, in a manner similar to the vertical light-colored portion **614** shown in FIG. 6B, but it will be apparent to those of skill in the art that, in other embodiments, the vertical light-colored portion **642** of the wireway **630** may extend along either of the sides **634** or **636**, in a manner similar to the vertical light-colored portions **612** and **616** shown in FIGS. 6A and 6C, respectively.

In some embodiments, an exemplary wireway **660**, as shown in perspective in FIG. 6E and in a front view in FIG. 6F, is substantially similar to the wireway **600** described above with reference to FIGS. 6A-6C, other than as described hereinafter. In some embodiments, the wireway

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660 has a flat top **662** and sides **664**, **666** that are substantially perpendicular to the flat top **662**. In some embodiments, the flat top **662** and angled sides **664**, **666** of the wireway **660** define a channel **668** within the wireway **660** to provide space to accommodate electrical cables. In some embodiments, the top surface of the wireway **660** (e.g., the side that faces away from the roof deck) includes at least one dark-colored portion and at least one light-colored portion. In some embodiments, the wireway **660** includes a horizontal light-colored portion **670** that is substantially similar to the horizontal light-colored portion **610** of the wireway **600** described above. In some embodiments, the wireway **660** includes a vertical light-colored portion **672**. The vertical light-colored portion **672** shown in FIG. 6E extends across the top surface **632** of the wireway **630** along the side, in a manner similar to the vertical light-colored portion **612** shown in FIG. 6A, but it will be apparent to those of skill in the art that, in other embodiments, the vertical light-colored portion **672** of the wireway **660** may extend along the side **636**, in a manner similar to the vertical light-colored portion **616** shown in FIG. 6C, or may extend across the top surface **632** intermediate the sides **634**, **636**, in a manner similar to the vertical light-colored portion **614** shown in FIG. 6B.

In some embodiments, an exemplary roofing system includes a masking element applied to the border between adjacent PV modules. FIG. 7 shows an exemplary masking element **700** applied to the border between adjacent PV modules **710**, **720** (e.g., the PV modules **510**, **520**, **530** as described above with reference to FIG. 5). In some embodiments, the masking element **700** is a dark color. In some embodiments, the masking element **700** is black. In some embodiments, the masking element **700** is made from a polymeric material. In some embodiments, the masking element **700** is made from a polymeric foam. In some embodiments, the masking element **700** is made from expanded polyethylene ("EPE"). In some embodiments, the masking element **700** is used in connection with the PV modules **710**, **720** that have a high-contrast backsheet **730**. In some embodiments, due to the high contrast of the backsheet **730**, any color difference between the masking element **700** and the PV modules **710**, **720** is swamped out, thereby hiding the boundary between the masking element **700** and the PV modules **710**, **720**. In some embodiments, by hiding the boundary between adjacent PV modules, use of the masking element **700** reduces the uniform, grid-like appearance of a roofing system including the PV modules **710**, **720**.

In some embodiments, a solar roofing system including one or more of the roofing shingle **300**, the PV module **510**, the wireway **600**, and/or the masking element **700** provides an appearance that includes similar degrees and types of irregular/randomized appearance across all elements of the roofing system. In some embodiments, such a randomized appearance is harmonized around the cell pitch, i.e., the sum of the width of each PV cell and the cell gap. In some embodiments, such a similarly-randomized appearance causes the various elements of the solar roofing system to visually blend with one another, thereby providing a more aesthetically pleasing appearance to the overall solar roofing system.

FIG. 8A shows an exemplary roofing system **800** that includes a plurality of the roofing shingle **300**, a plurality of the PV module **510**, a plurality of the wireway **600**, and a plurality of the masking element **700**. For clarity, only one of each of the elements noted above is specifically identified by a reference numeral in FIG. 8A. FIG. 8B shows a magnified view of a portion of the roofing system **800**

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including a plurality of the roofing shingle 300, a plurality of the PV module 510, and a plurality of the wireway 600 separating adjacent ones of the PV module 510. FIG. 8C shows a magnified view of a portion of the roofing system 800 including a plurality of the roofing shingle 300, a plurality of the PV module 510, and a plurality of the masking element 700 separating adjacent ones of the PV module 510 from adjacent ones of the roofing shingle 300. FIG. 8D shows a magnified view of a portion of the roofing system 800 including a plurality of the roofing shingle 300, a plurality of the PV module 510, and a plurality of the wireway 600 separating ones of the PV module 510 from adjacent ones of the roofing shingle 300. FIG. 8E shows a magnified view of a portion of the roofing system 800 including a plurality of the roofing shingle 300, a plurality of the PV module 510, and a plurality of the masking element 700 separating adjacent ones of the PV module 510. It may be seen from FIGS. 8A-8E that the roofing system 800 including a plurality of the roofing shingle 300, a plurality of the PV module 510, a plurality of the wireway 600, and a plurality of the masking element 700 provides an aesthetic appearance in which the various elements of the roofing system 800 blend with one another aesthetically, rather than one in which the PV module 510 stands out. It will be apparent to those of skill in the art that the roofing system 800 shown in FIGS. 8A-8E is only one exemplary manner of arranging a plurality of the roofing shingle 300, a plurality of the PV module 510, a plurality of the wireway 600, and a plurality of the masking element 700, and that any number of other arrangements of these same elements may be made. It will be further apparent to those of skill in the art that while the roofing system 800 shown in FIGS. 8A-8E includes all of the roofing shingle 300, the PV module 510, the wireway 600, and the masking element 700, the same or similar aesthetic effect may be accomplished with a subset of these elements (including, but not limited to, with the roofing shingle 300 and the PV module 510; with the roofing shingle 300, the PV module 510, and the wireway 600; or with the roofing shingle 300, the PV module 510, and the masking element 700).

In some embodiments, the various elements of the exemplary roofing system 800 mimic the water-shedding ability of a conventional roof shingle. In some embodiments, the various elements of the exemplary roofing system can be affixed to a roof deck using typical roofing methods such as nails or screws.

In some embodiments, the roofing system 800 also includes at least one starter bar, a foot module, and a plurality of water shedding layers. In some embodiments, the roofing shingle 300 and/or the PV module includes an upper portion and a lower portion and is configured to be installed such that the upper portion is at a higher elevation than the lower portion. In some embodiments, the at least one starter bar is configured to be installed to a roof deck and includes a foot base. In some embodiments, a first one of the water shedding layers is configured to be installed over the foot base of the at least one starter bar, and at least one other one of the water shedding layers is configured to overlap and be installed over the first one of the plurality of water shedding layers. In some embodiments, the foot module is configured to be attached to the upper portion of the PV module 510 and/or the roofing shingle 300. In some embodiments, the lower portion of the PV module 510 and/or the roofing shingle 300 is adapted to align with the foot base of the at least one starter bar, and the foot module is configured

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to be affixed to a last overlapping layer of the at least one of another of the first plurality of water shedding layers to the roof deck.

While a number of embodiments of the present invention have been described, it is understood that these embodiments are illustrative only, and not restrictive, and that many modifications may become apparent to those of ordinary skill in the art. Further still, the various steps may be carried out in any desired order (and any desired steps may be added and/or any desired steps may be eliminated).

What is claimed is:

1. A roofing system, comprising:

- a plurality of photovoltaic modules on a roof deck, wherein each of the plurality of photovoltaic modules includes a plurality of photovoltaic cells, wherein each of the plurality of photovoltaic cells has one constant photovoltaic cell width; and
- a plurality of roofing shingles proximate to the plurality of photovoltaic modules on the roof deck, wherein each of at least some of the plurality of roofing shingles having
 - a top surface,
 - an exposure zone at a lower end of the top surface and
 - a headlap zone at an upper end of the top surface, and
 - a plurality of tooth portions extending from the lower end toward the headlap zone, wherein a first one of the plurality of tooth portions has a first width, wherein the first width is the photovoltaic cell width,
 - wherein a second one of the plurality of tooth portions has a second width, wherein the second width is the photovoltaic cell width multiplied by a first positive integer that is greater than 1,
 - wherein a third one of the plurality of tooth portions has a third width, wherein the third width is the photovoltaic cell width multiplied by a second positive integer that is greater than 1 and different than the first positive integer,
 - wherein each of the roofing shingles does not include a photovoltaic cell.

2. The roofing system of claim 1, wherein the plurality of roofing shingles comprises at least a first roofing shingle, wherein the first roofing shingle comprises thermoplastic olefin, polyvinyl chloride, or asphalt.

3. The roofing system of claim 2, wherein a top surface of the first roofing shingle comprises embedded granules.

4. The roofing system of claim 1, wherein the plurality of roofing shingle comprises at least a first roofing shingle and a second roofing shingle, wherein an arrangement of tooth portions of the second roofing shingle is not identical to an arrangement of tooth portions of the first roofing shingle.

5. The roofing system of claim 1, wherein the plurality of roofing shingle comprises at least a first roofing shingle and a second roofing shingle, wherein an arrangement of tooth portions of the second roofing shingle is identical to an arrangement of tooth portions of the first roofing shingle.

6. The roofing system of claim 1, wherein each of the first and second positive integers is selected from the group consisting of 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, and 20.

7. The roofing system of claim 1, wherein the plurality of tooth portions of at least some of the plurality of roofing shingles comprises a fourth one of the plurality of tooth

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portions having a fourth width, wherein the fourth width is the photovoltaic cell width multiplied by 0.5.

8. A roofing system, comprising:

a plurality of photovoltaic modules on a roof deck,
 wherein each of the plurality of photovoltaic modules
 includes a plurality of photovoltaic cells,
 wherein each of the plurality of photovoltaic cells
 has one constant photovoltaic cell width; and
 a plurality of roofing shingles proximate to the plurality of
 photovoltaic modules on the roof deck,
 wherein each of at least some of the plurality of roofing
 shingles having
 a top surface,
 an exposure zone at a lower end of the top surface
 and
 a headlap zone at an upper end of the top surface, and
 a plurality of tooth portions extending from the lower
 end toward the headlap zone,
 wherein a first one of the plurality of tooth por-
 tions has a first width,
 wherein the first width is the photovoltaic cell
 width multiplied by a first positive integer that
 is greater than 1,
 wherein a second one of the plurality of tooth
 portions has a second width,
 wherein the second width is the photovoltaic cell
 width multiplied by a second positive integer
 that is greater than 1 and different than the first
 positive integer,
 wherein each of the roofing shingles does not include a
 photovoltaic cell.

9. The roofing system of claim 8, wherein the plurality of roofing shingles comprises at least a first roofing shingle, wherein the first roofing shingle comprises thermoplastic olefin, polyvinyl chloride, or asphalt.

10. The roofing system of claim 9, wherein a top surface of the first roofing shingle comprises embedded granules.

11. The roofing system of claim 8, wherein the plurality of roofing shingle comprises at least a first roofing shingle and a second roofing shingle, wherein an arrangement of tooth portions of the second roofing shingle is not identical to an arrangement of tooth portions of the first roofing shingle.

12. The roofing system of claim 8, wherein the plurality of roofing shingle comprises at least a first roofing shingle and a second roofing shingle, wherein an arrangement of tooth portions of the second roofing shingle is identical to an arrangement of tooth portions of the first roofing shingle.

13. The roofing system of claim 8, wherein each of the first and second positive integers is selected from the group consisting of 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, and 20.

14. The roofing system of claim 8, wherein the plurality of tooth portions of at least some of the plurality of roofing shingles comprises a third one of the plurality of tooth portions having a third width, wherein the third width is the photovoltaic cell width multiplied by 0.5.

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15. A method, comprising:

obtaining a plurality of photovoltaic modules,
 wherein each of the plurality of photovoltaic modules
 includes a plurality of photovoltaic cells,
 wherein each of the plurality of photovoltaic cells
 has one constant photovoltaic cell width; and
 obtaining a plurality of roofing shingles,
 wherein each of at least some of the plurality of roofing
 shingles having
 a top surface,
 an exposure zone at a lower end of the top surface
 and
 a headlap zone at an upper end of the top surface, and
 a plurality of tooth portions from the lower end
 toward the headlap zone,
 wherein a first one of the plurality of tooth por-
 tions has a first width,
 wherein the first width is the photovoltaic cell
 width,
 wherein a second one of the plurality of tooth
 portions has a second width,
 wherein the second width is the photovoltaic
 cell width multiplied by a first positive integer
 that is greater than 1,
 wherein a third one of the plurality of tooth
 portions has a third width,
 wherein the third width is the photovoltaic cell
 width multiplied by a second positive integer
 that is greater than 1 and different than the first
 positive integer,
 wherein each of the roofing shingles do not include a
 photovoltaic cell; and
 installing the plurality of photovoltaic modules and the
 plurality of roofing shingles proximate one another
 above a roof deck.

16. The method of claim 15, wherein the plurality of roofing shingles comprises at least a first roofing shingle, wherein the first roofing shingle comprises thermoplastic olefin, polyvinyl chloride, or asphalt.

17. The method of claim 15, wherein a top surface of the first roofing shingle comprises embedded granules.

18. The method of claim 15, wherein each of the first and second positive integers is selected from the group consisting of 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, and 20.

19. The method of claim 15, wherein the plurality of tooth portions of at least some of the plurality of roofing shingles comprises a fourth one of the plurality of tooth portions having a fourth width, wherein the fourth width is the photovoltaic cell width multiplied by 0.5.

20. The method of claim 19, wherein each of the first and second positive integers is selected from the group consisting of 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, and 20.

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